

*Part B*Question 9 from section B **Question 1:**

$$B_d((0, 0), 1), B_d[(0, 0), 1], \text{ and } S_s((0, 0), 1)$$

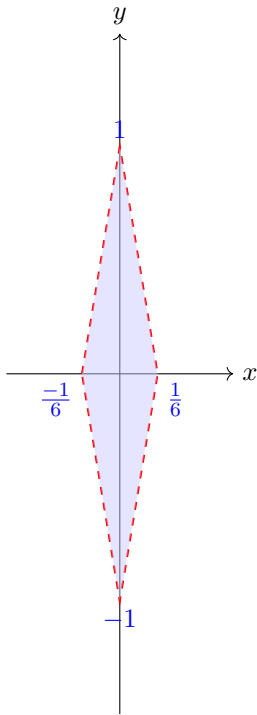
a)

The open ball on metric

$$d((a_1, a_2), (b_1, b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$$

is given by

$$\begin{aligned} B_d((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : 6|x - 0| + |y - 0| < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : 6|x| + |y| < 1\} \end{aligned}$$

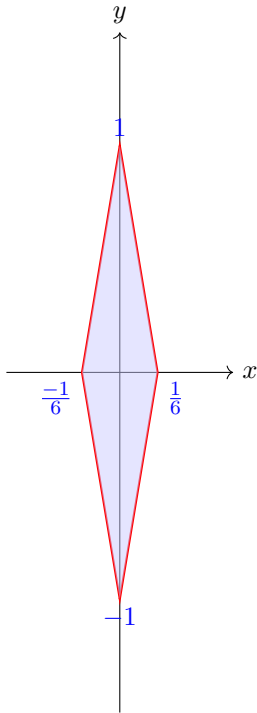


The closed ball on metric

$$d((a_1, a_2), (b_1, b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$$

is given by

$$\begin{aligned}
 B_d[(0,0),1] &= \{(x,y) \in \mathbb{R}^2 : d((x,y), (0,0)) \leq 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x-0| + |y-0| \leq 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x| + |y| \leq 1\}
 \end{aligned}$$

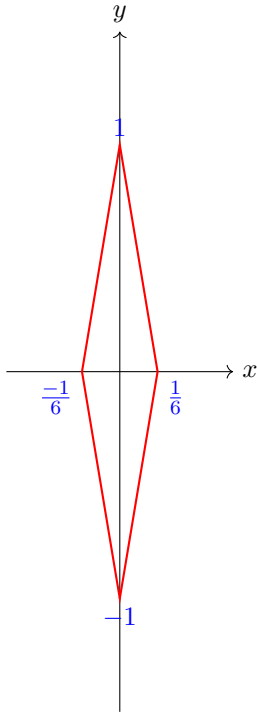


The sphere on metric

$$d((a_1, a_2), (b_1, b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$$

is given by

$$\begin{aligned}
 S_s((0,0),1) &= \{(x,y) \in \mathbb{R}^2 : d((x,y), (0,0)) = 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x-0| + |y-0| = 1\} \\
 &= \{(x,y) \in \mathbb{R}^2 : 6|x| + |y| = 1\}
 \end{aligned}$$



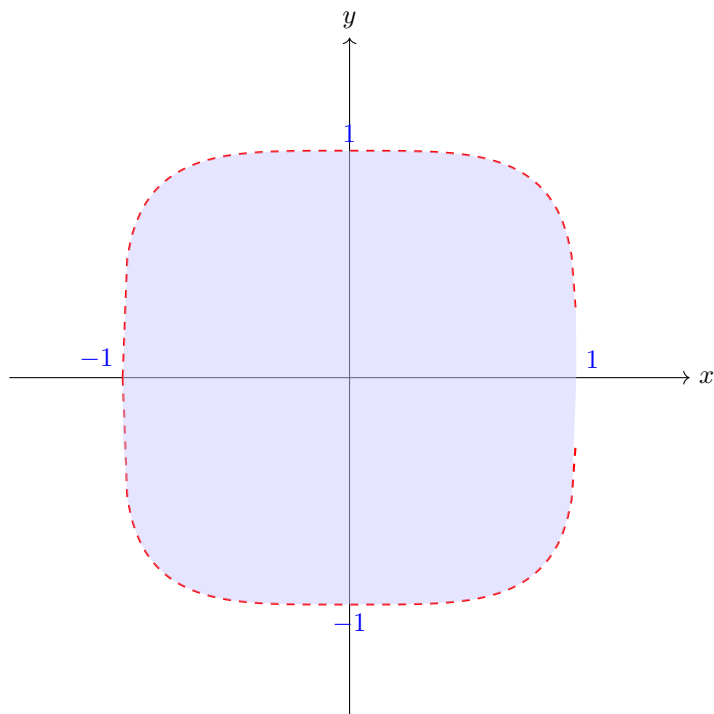
b)

The open ball on metric

$$d((a_1, a_2), (b_1, b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{\frac{1}{4}}$$

is given by

$$\begin{aligned} B_d[(0, 0), 1] &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x - 0|^4 + |y - 0|^4)^{\frac{1}{4}} < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x|^4 + |y|^4)^{\frac{1}{4}} < 1\} \end{aligned}$$

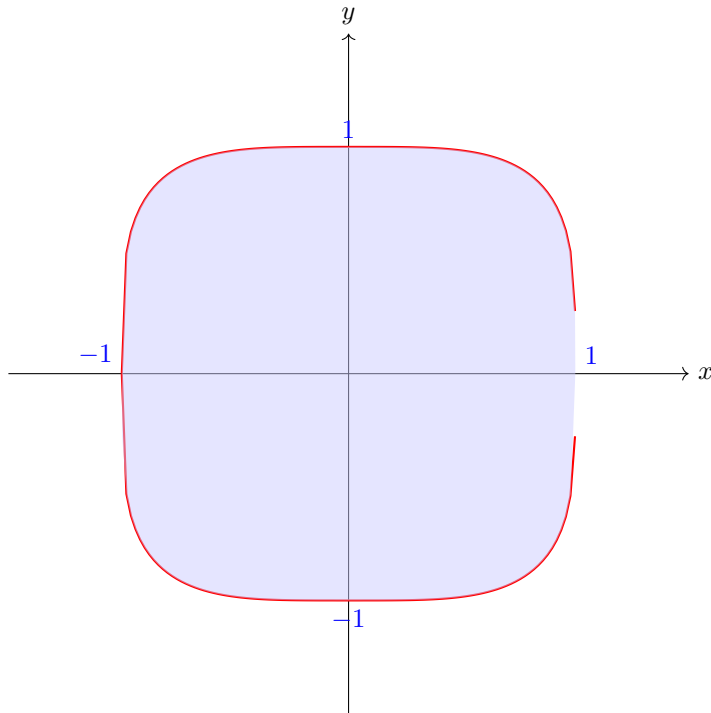


The closed ball on metric

$$d((a_1, a_2), (b_1, b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{\frac{1}{4}}$$

is given by

$$\begin{aligned} B_d[(0,0), 1] &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x - 0|^4 + |y - 0|^4)^{\frac{1}{4}} \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x|^4 + |y|^4)^{\frac{1}{4}} \leq 1\} \end{aligned}$$

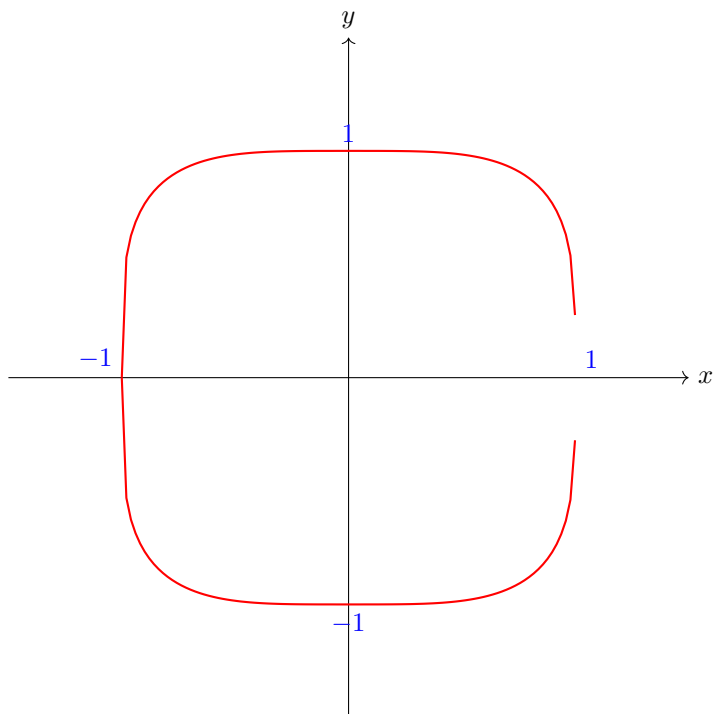


The sphere on metric

$$d((a_1, a_2), (b_1, b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{\frac{1}{4}}$$

is given by

$$\begin{aligned} S_s((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) = 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x - 0|^4 + |y - 0|^4)^{\frac{1}{4}} = 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : (|x|^4 + |y|^4)^{\frac{1}{4}} = 1\} \end{aligned}$$



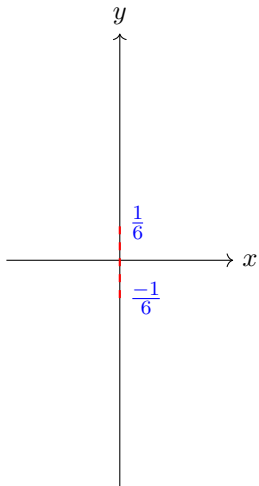
c)

The open ball on metric

$$d((a_1, a_2), (b_1, b_2)) = d_0(a_1, b_1) + 6|b_2 - a_2|$$

is given by

$$\begin{aligned} S_s((0, 0), 1) &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y - 0| < 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y| < 1\} \end{aligned}$$

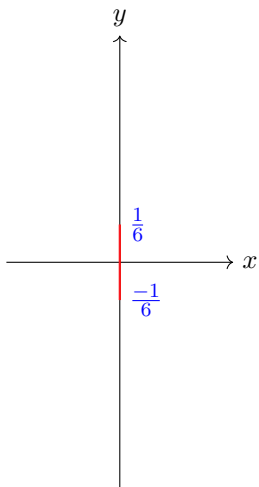


The closed ball on metric

$$d((a_1, a_2), (b_1, b_2)) = d_0(a_1, b_1) + 6|b_2 - a_2|$$

is given by

$$\begin{aligned} B_d[(0, 0), 1] &= \{(x, y) \in \mathbb{R}^2 : d((x, y), (0, 0)) \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y - 0| \leq 1\} \\ &= \{(x, y) \in \mathbb{R}^2 : d_0(x, 0) + 6|y| \leq 1\} \end{aligned}$$



The sphere on metric

$$d((a_1, a_2), (b_1, b_2)) = d_0(a_1, b_1) + 6|b_2 - a_2|$$

is given by

$$\begin{aligned} S_s((0,0),1) &= \{(x,y) \in \mathbb{R}^2 : d((x,y), (0,0)) = 1\} \\ &= \{(x,y) \in \mathbb{R}^2 : d_0(x,0) + 6|y-0| = 1\} \\ &= \{(x,y) \in \mathbb{R}^2 : d_0(x,0) + 6|y| = 1\} \end{aligned}$$

